

Mathematics A

Unit 2 • Chapter OL-T20 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 3 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T20 • Expertise Q16+ • 2 items • 3 marks

Chapter	Items	Marks	Starts
01. OL-T20 Completing the Square	2	3	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Completing the Square

Unit 2 • Expertise • 3 marks • 2 item(s)

November 2021 | 4MA1-1H Q19 | 2 marks

Negative leading coefficient or maximum form

November 2021 | 4MA1-1H Q19 | 1 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

19 $ABCED$ is a five-sided shape.

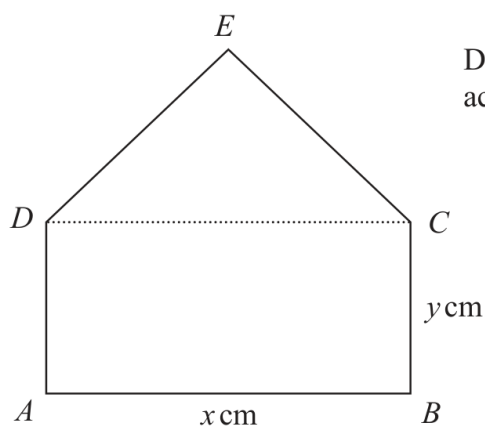


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

.....

.....

.....

(1)

Worked solution

November 2021 | Question 19 | 2 marks

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method / working

1. The perimeter is made from three sides of length $\sqrt{x}$ and two sides of length $\sqrt{y}$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $3x+2y=100$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $y=\frac{100-3x}{2}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The area is the area of the rectangle plus the area of the equilateral triangle:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $R=xy+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. Substitute $y=\frac{100-3x}{2}$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $R=x\left(\frac{100-3x}{2}\right)+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $R=50x-\frac{3}{2}x^2+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. For the maximum, write

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $R=-\frac{6-\sqrt{3}}{4}x^2+50x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. The maximum occurs at

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $x=\frac{-b}{2a}=\frac{-50}{2\left(-\frac{6-\sqrt{3}}{4}\right)}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $x=\frac{100}{6-\sqrt{3}}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. The coefficient of (x^2) is negative, so the quadratic graph opens downwards and this vertex gives a maximum.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. Answer: $\left(R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)\right)$, and $\left(x=\frac{100}{6-\sqrt{3}}\right)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$, and $x=\frac{100}{6-\sqrt{3}}$.

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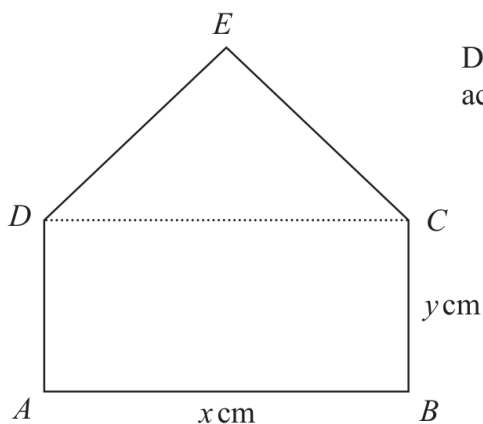


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November 2021 | Question 19 | 1 marks

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Final answer: $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$, and $x=\frac{100}{6-\sqrt{3}}$.